Name: Sam, Sophia, Kyle, Ryan **Date:** 04-11-24 **Class:** E4USA

MyDesign Portfolio

Element G Initial Template

How to use this template

Teachers: You may customize this document before providing it to students.

Students: The template below is one way to present your construction of a testable prototype, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit for your own prototype construction details, be sure to delete the instructions.

Guidance before filling out the template that follows:

- ➔ Because you will iterate on your design multiple times, you will need to create multiple copies of these sections. Think of this Element as another piece of a diary that you will date and keep adding to throughout your work until your final iteration. It's critical that iteration shines through in this Element also.
- ➔ You will then proceed on to test and analyze this design and then determine your next iteration, returning to this Element afterwards when you are ready to construct a new testable prototype. Fill out each of these steps again.

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation *(G1)*

This section focuses on the process of explaining your prototypes. While the title of the subelement focuses on your final prototype, you must have at least an initial prototype and a final prototype, with additional prototypes in-between being desirable. Note that you never know which prototype is your final prototype until after you test it, so be sure to record all of the information for each prototype. Use as many copies of this sub-template as needed.

Ensure that photos are taken from multiple angles so that all sides and subsystems can be seen. All images should have captions and should be referenced in-text.

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Prototype Construction of Design #1 (make as many copies, updating the design number, as needed)

Date:
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:
Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

Steps	Details	Tools and Manufacturing Methods Used
Step 1	Build the activated carbon filter, creating a square frame of popsicle sticks and filling in the space with black construction paper.	Popsicle sticks, black construction paper, tape
Step 2	Build the fan using three cut popsicle sticks as blades, held together by construction paper.	Popsicle sticks, construction paper, glue
Step 3	Create a fan compartment using popsicle sticks. Attach the fan apparatus to the fan compartment.	Popsicle sticks, tape, fan apparatus
Step 4	Build frames for the sides of the air purifier using popsicle sticks on the edges with paper as the surface of the sides	Popsicle sticks, tape, construction paper
Step 5	Create the filter compartment by leaving a space (a popsicle stick wide) between the fan compartment and the rest of the apparatus, taping them together. Insert the constructed air filter in the space.	Popsicle sticks, tape, built frame, fan compartment

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Step 6	Cut a hole in the front paper covering to create the small-area opening.	scissors
Step 7	Draw the electric chord (power source) on the side of the filter using a pen.	pen

Prototype Construction of Design #2 (make as many copies, updating the design number, as needed)

Date:
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:
Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

Steps	**Details**	**Tools and Manufacturing Methods Used**
Step 1	Start our prototype. We first measured our plexiglass to create a box shape	Ruler, sharpie, plexiglass
Step 2	Score our plexiglass to ensure an easy cut.	X-acto knife
Step 3	Clamp plexiglass to the table and use the wire saw to cut our plexiglass.	Wire saw, plexiglass, clamps
Step 4	Cut an opening in the back piece of plexiglass to allow for quicker airflow into our filter.	Wire saw, plexiglass, clamp
Step 5	Create our filterbox by cutting cardboard to fit our filters. Then, affixing the metal grate to the	Hotglue, cardboard, scissors, metal grate

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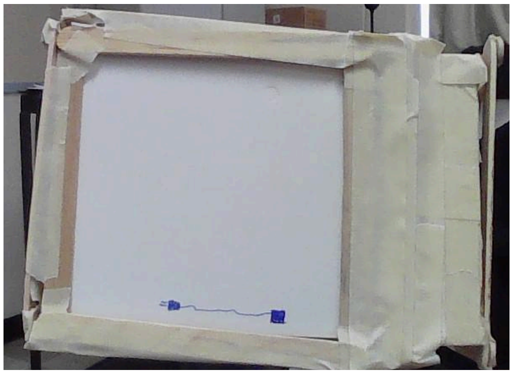
	cardboard in order to allow airflow.	
Step 6	Glue plexiglass into a box shape.	Hot glue, plexiglass
Step 7	Add our fan and glue it in.	Fan, hotglue
Step 8	Attach the filter box into our main filter box, and then add the top of our filter.	Hot glue, plexiglass, cardboard.

Scale Model:

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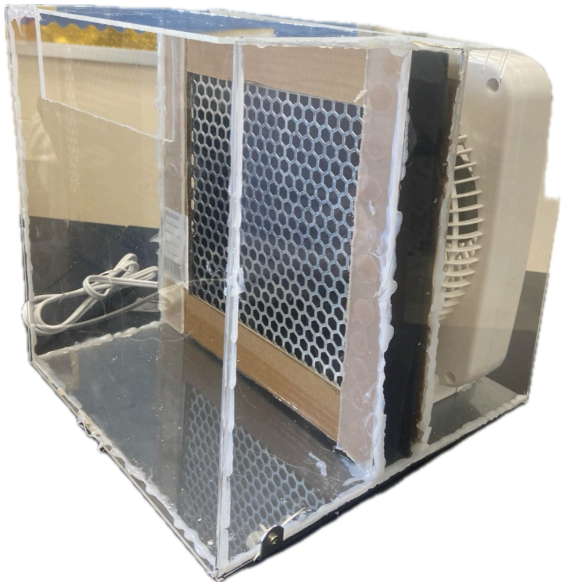
Prototype:



Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #3 (change this number to match the design number you're on)

Date:
Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:
Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

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Steps	Details	Tools and Manufacturing Methods Used
Step 1	Remove the hot glue from our plexiglass box and reattach the plexiglass with silicone.	Silicone, exacto knife
Step 2	Add rubber stoppers to the bottom of our box.	Rubber stoppers, hotglue
Step 3	Attach a plastic hinge to the top of our filter box.	Hot glue, hinge
Step 4		
Step 5		
Step 6		
Step 7		

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

List of material details, including item name, description, where it was purchased/obtained (include URL if applicable), cost per unit (if applicable), # of units, total cost (if applicable)

Material name	Description	Where material was purchased	Cost per unit	# units	Total cost per item

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Fan	Comfort Zone Box Fan, 9 inch, Portable, Electric Quiet, 3 Speed, Small Box Fan, Desk Fan, Table Fan, Airflow 9.65 ft/sec	Amazon	$ 14.99	1	$ 14.99
Activated Carbon Filter	Anti-Odor Material, Activated Charcoal Carbon Air Filter for Air Purifiers, Vent Hoods, Vacuums, Aquariums, Microwaves	Amazon	$ 15.99	1	$ 15.99
PlexiGlass	Acrylic Sheets - Clear Acrylic Sheet Set for Displays - 2 Pack Plexiglass Sheets for Signs - 1/8" Thick (3mm) Cast	Amazon	$ 11.99	4	$ 47.96

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	Plexiglass Sheet for DIY Projects				
Screws/Brackets	Bracket Corner Brace MONKIPAER Metal Corner bracket 90 Degree Angle Stainless Steel Bracket with 100 Pcs Screws for securing wooden frames tables chairs bed furniture and other DIY structural	Amazon	$ 6.85	1	$ 6.85
Filter Grate	Dryer Vent Grill Dryer Vent Screen with Aluminum Built Bird Guards Inserts Stop Variety Birds Nesting in Dryer Vents and	Amazon	$ 7.59	1	$ 7.59

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	Bathroom Exhaust Vents Outdoor, 3 Inch to 8 Inch				
Silicone	https://www.amazon.com/Loctite-Silicone-Waterproof-2-7-Ounce-908570/dp/B0002BBX3U?source=ps-sl-shoppingads-lpcontext&ref_=fplfs&smid=ATVPDKIKX0DER&th=1	Amazon	$5.50	3	$19.50
Hinges	https://www.amazon.com/Acrylic-Adhesive-Transparent-Continuous	Amazon	$7.00	1	$7.00
Stoppers	https://www.amazon.com/Scotch-Bumpers-Pack-Clear-SP951-NA/dp/B01N74P0QB/ref=sr_1_3?	Amazon	$5.50	1	$5.50

Detailed sketches, which might include CAD if you have learned it in your class:

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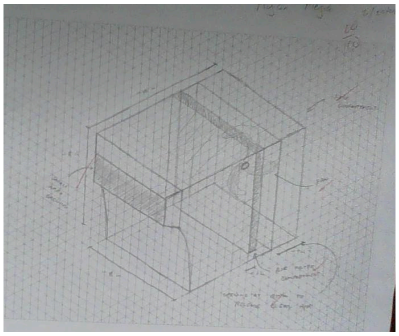

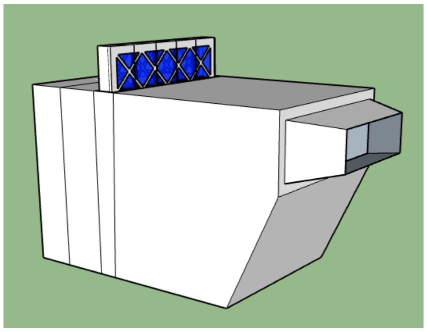

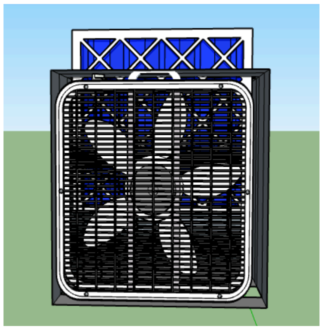

Construction of the Final Prototype *(G2)*

When you've finished construction of your final prototype, you must be sure that your prototype allows sufficient functionality that you could collect objective, measurable data for all or nearly all of your design requirements. This requires you to look back at your design requirements from Element C. Copy all of your design requirements from Element C into this table and then add how your prototype will allow you to collect objective, measurable data. Note any weaknesses you find or details that will require expert review because they cannot be mathematically modeled or tested,

For example, if you have a design requirement that your design can "transport medicine a distance of 5 meters in 1 minute", is your prototype sufficiently functional such that you can test that? (e.g. can the prototype move the medicine at least a little bit?)

Design Requirement	How will this prototype allow you to collect objective, measurable data?
Reduce VOC	We will measure VOC levels in a simulation in the fume hood. We need a range between 0 and 400 ppb.
Space efficient	Measure the apparatus. Our filter is 11"x11"x12"
Reduce HCHO	We will measure HCHO in the fume hood. We need a range between 0.5-2.0 ppm
Reduce PM 1	Need a range between 17.5 μg/m3 and 85.0 μg/m3.

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Reduce PM 2.5	Need a range between 35 µg/m3 and 75 µg/m3.

Testing or Modeling of Attributes *(G3, G4)* and Addressing Non-Testable Attributes *(G4)*

Describe all of your prototype's attributes (features, aspects, details, or subsystems). Demonstrate that your prototype is constructed with sufficient functionality that your prototype can test all attributes of your design. This requires you to look back at your initial proposed design details from Element D.

For example, if your initial design details called for wheels and motors, is your prototype sufficiently functional such that you can test out those wheels and motors?

If there are aspects of your design that you believe cannot be physically prototyped or mathematically modeled within the scope and limits of your class, you must later provide evidence that an expert has reviewed your justifications and that you have incorporated their feedback. You do not need to conduct that review in Element G. For now, enter "expert review required" for these attributes in the final column.

Attribute	Description	Describe your prototype's functionality to test this attribute
Box Fan	Power source; Pulls air in through the apparatus from the back and returns filtered air to the classroom.	Will be able to be turned on and off while in testing; air is actively pulled from the front opening (use paper).
Activated Carbon Filter	Absorbs particulate matter	Test air quality values

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	and odors (including VOCs) via lodging in pores.	before and after the class in which it is being used
Small-Area Opening	Smaller area allows for quicker airflow, based on the physics principle of Bernoulli's law, resulting in quicker filtration.	Air is felt being actively pulled through the opening by placing a paper in front of the opening.
Filter Compartment	Latched top door allows for simple swap of activated carbon filters.	Compartment should be easily opened/closed; filter easily taken out/inserted by student.
Minimal Dimensions	Must be small enough to fit in the space we were given (15x25x10 ft) and be mounted to the wall or hung from the ceiling.	Measure filter's cubic shape; find volume by measuring length, width, and height.
VOC, HCHO, PM 1, PM 2.5 Filtration	Filter must result in a decrease of these elements in the classroom.	To be measured by the instructor's multi-purpose instrument.
ABS Filtration	Filter must lessen odor of ABS plastics in the classroom.	Filtration will be determined by presence (or lack thereof) of odor alone.
Student Safety	Apparatus must be safe while in operation and while students change the filter.	N/A
Durability	Apparatus must have a significantly long lifespan.	(Cannot be determined in one testing phase)

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Addressing Non-Testable Attributes *(G4)*

For all attributes that you listed "expert review required" in the table above, provide a justification for why that is true. "Field experts" are professionals who spend at least a portion of their time working in a field directly related to your project. These field experts are not permitted to double count as stakeholders.

Attribute	Justification for why this attribute cannot be modeled or tested with your prototype
Student Safety and Space Effectiveness	Cannot be qualitatively measured; Will be based on observation during standard operation and during the filter replacement process carried out by a student.
Durability	Long lifespan can only be determined after several uses beyond our provided testing times; Apparatus will not undergo stress tests.

Test Results

Test Results	
Original	
PM2.5	802
PM1	326
HCHO	0.021
TVOC	0.121
Final (after 6 minutes of filter use)	

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PM2.5	081
PM1	014
HCHO	0.014
TVOC	0.081

(particulate matter reached baseline at 4:09 mins)

(L to R: Baseline, Initial Value, Final Value after 6 mins)